# Chapter 4. Constrained Optimization (I) : P1-2

## Problem 1

(a) We first note that the given constraint set is not bounded as $x = 3/z$ and $z$ can approach 0. (See Figure 1 below)

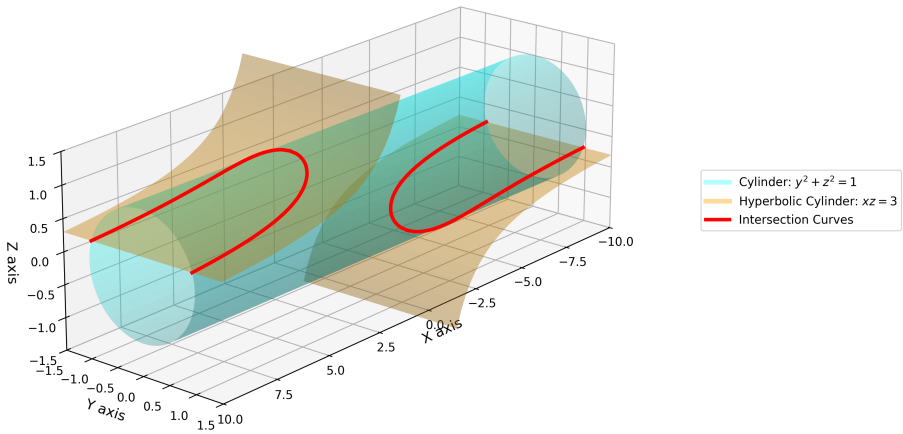


Figure 1: The constraint set

(i) The Lagrangian function is

$$L(x, y, z, \mu_1, \mu_2) = yz - xz - \mu_1((y^2 + z^2) - 1) - \mu_2(xz - 3).$$

The FOC are given by

$$\begin{cases} L_x = 0 \\ L_y = 0 \\ L_z = 0 \\ L_{\mu_1} = 0 \\ L_{\mu_2} = 0 \end{cases} \Rightarrow \begin{cases} -z - \mu_2 z = 0, & \cdots (1) \\ z - \mu_1(2y) = 0, & \cdots (2) \\ y - x - \mu_1(2z) - \mu_2 x = 0, & \cdots (3) \\ y^2 + z^2 = 1, & \cdots (4) \\ xz = 3, & \cdots (5) \end{cases}$$

(ii) Let $\mathbf{p} = (x, y, z)$ be any point from the constraint set. Then its Jacobian matrix is

$$D\mathbf{h}(\mathbf{p}) = \begin{pmatrix} 0 & 2y & 2z \\ z & 0 & x \end{pmatrix}.$$

Since $xz = 3$, $x \neq 0$ and $z \neq 0$, the second row is non-zero. The first row has first component 0, while the second row has first component $z = 0$. Hence, the second row cannot be a scalar multiple of the first row. Then the two rows are linearly independent. Therefore, the Jacobian matrix is always of rank 2 on the constraint set which means that NDCQ is satisfied for this problem.

(iii) Firstly, by (5), we have $z \neq 0$. Hence by (1), we have

$$\mu_2 = -1.$$

Then (3) becomes

$$y - x - \mu_1(2z) + x = 0.$$

Thus,

$$y = 2\mu_1 z.$$

Also, (2) gives

$$z - \mu_1(2y) = 0.$$

Putting $y = 2\mu_1 z$ into (2), we have

$$z - \mu_1(2)(2\mu_1 z) = 0.$$

Hence,

$$z - 4\mu_1^2 z = 0.$$

Since $z \neq 0$, we have

$$1 - 4\mu_1^2 = 0.$$

Therefore,

$$\mu_1 = \frac{1}{2} \quad \text{or} \quad \mu_1 = -\frac{1}{2}.$$

- If $\mu_1 = \dfrac{1}{2}$, then

$$y = 2\left(\frac{1}{2}\right) z = z.$$

Putting this into (4), we have

$$z^2 + z^2 = 1.$$

Hence,

$$2z^2 = 1, \qquad z = \pm\frac{1}{\sqrt{2}}.$$

By (5),

$$x = \frac{3}{z}.$$

Therefore,

$$(x, y, z, \mu_1, \mu_2) = \left(3\sqrt{2}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{2}, -1\right), \quad \left(-3\sqrt{2}, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, \frac{1}{2}, -1\right),$$

at which $f = -\frac{5}{2}$.

- If $\mu_1 = -\dfrac{1}{2}$, then

$$y = 2\left(-\frac{1}{2}\right)z = -z.$$

Putting this into (4), we have

$$(-z)^2 + z^2 = 1.$$

Hence,

$$2z^2 = 1, \qquad z = \pm\frac{1}{\sqrt{2}}.$$

By (5),

$$x = \frac{3}{z}.$$

Therefore,

$$(x, y, z, \mu_1, \mu_2) = \left(3\sqrt{2}, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, -\frac{1}{2}, -1\right), \quad \left(-3\sqrt{2}, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, -\frac{1}{2}, -1\right),$$

at which $f = -\frac{7}{2}$.

Thus, we obtain four critical points

$$(x, y, z, \mu_1, \mu_2) = \left(3\sqrt{2}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{2}, -1\right), \ \left(-3\sqrt{2}, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, \frac{1}{2}, -1\right),$$

$$\left(3\sqrt{2}, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, -\frac{1}{2}, -1\right), \left(-3\sqrt{2}, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, -\frac{1}{2}, -1\right).$$

As the constraint set are not bounded, we need to further determine if the values of $f$ in the critical points are indeed the global maximum.

Since $xz = 3$, we have

$$f(x, y, z) = yz - xz = yz - 3.$$

Also, from the constraint

$$y^2 + z^2 = 1,$$

we have

$$yz \leq \frac{y^2 + z^2}{2} = \frac{1}{2}.$$

Hence,

$$f(x, y, z) = yz - 3 \leq \frac{1}{2} - 3 = -\frac{5}{2}.$$

Equality occurs when

$$y = z = \pm\frac{1}{\sqrt{2}}.$$

Therefore, the maximum value of $f$ is $-\frac{5}{2}$.

(b) We first note that the given constraint set is not bounded. (See Figure 2 below)

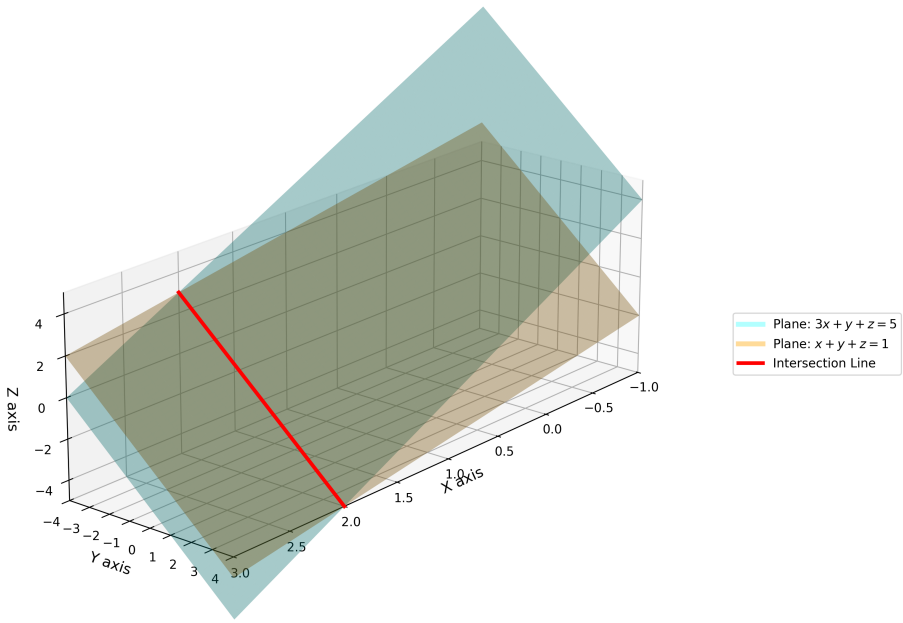


Figure 2: The constrain set

(i) The Lagrangian function is

$$L(x, y, z, \mu_1, \mu_2) = x^2 + y^2 + z^2 - \mu_1((3x + y + z) - 5) - \mu_2((x + y + z) - 1).$$

The FOC are given by

$$\begin{cases} L_x = 0 \\ L_y = 0 \\ L_z = 0 \\ L_{\mu_1} = 0 \\ L_{\mu_2} = 0 \end{cases} \Rightarrow \begin{cases} 2x - 3\mu_1 - \mu_2 = 0, & \cdots (1) \\ 2y - \mu_1 - \mu_2 = 0, & \cdots (2) \\ 2z - \mu_1 - \mu_2 = 0, & \cdots (3) \\ 3x + y + z = 5, & \cdots (4) \\ x + y + z = 1, & \cdots (5) \end{cases}$$

(ii) Let $\mathbf{p} = (x, y, z)$ be any point from the constraint set. Then its Jacobian matrix is

$$D\mathbf{h}(\mathbf{p}) = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \xrightarrow{\text{REF}} \begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & -2 \end{pmatrix}$$

which is of rank 2. Therefore, $\mathbf{p}$ satisfies the NDCQ. Moreover, since $\mathbf{p}$ is

arbitrary, we conclude that NDCQ is valid for this constraint set.

(iii) Firstly, by subtracting (5) from (4), we have

$$(3x + y + z) - (x + y + z) = 5 - 1.$$

Hence,

$$2x = 4, \qquad x = 2.$$

Then by (5), we have

$$2 + y + z = 1.$$

Thus,

$$y + z = -1.$$

Next, subtracting (3) from (2), we have

$$(2y - \mu_1 - \mu_2) - (2z - \mu_1 - \mu_2) = 0.$$

Hence,

$$2y - 2z = 0, \qquad y = z.$$

Putting $y = z$ into $y + z = -1$, we have

$$2y = -1.$$

Therefore,

$$y = z = -\frac{1}{2}.$$

Now, by (2), we have

$$2\left(-\frac{1}{2}\right) - \mu_1 - \mu_2 = 0.$$

Hence,

$$\mu_1 + \mu_2 = -1.$$

Also, by (1), we have

$$2(2) - 3\mu_1 - \mu_2 = 0.$$

Hence,

$$3\mu_1 + \mu_2 = 4.$$

Solving

$$\begin{cases} \mu_1 + \mu_2 = -1, \\ 3\mu_1 + \mu_2 = 4, \end{cases}$$

we obtain

$$\mu_1 = \frac{5}{2}, \qquad \mu_2 = -\frac{7}{2}.$$

Thus, we obtain one critical point

$$(x, y, z, \mu_1, \mu_2) = \left(2, -\frac{1}{2}, -\frac{1}{2}, \frac{5}{2}, -\frac{7}{2}\right),$$

at which $f = \frac{9}{2}$.

Since the constraints imply

$$x = 2, \qquad y + z = -1,$$

we have

$$f(x, y, z) = x^2 + y^2 + z^2 = 4 + y^2 + z^2.$$

By

$$y^2 + z^2 \geq \frac{(y+z)^2}{2},$$

we get

$$y^2 + z^2 \geq \frac{(-1)^2}{2} = \frac{1}{2}.$$

Therefore,

$$f(x, y, z) \geq 4 + \frac{1}{2} = \frac{9}{2}.$$

Equality occurs when

$$y = z = -\frac{1}{2}.$$

Hence, the minimum value of $f$ is $\frac{9}{2}$.

(c) We first note that the given constraint set is not bounded. (See Figure 3 below)

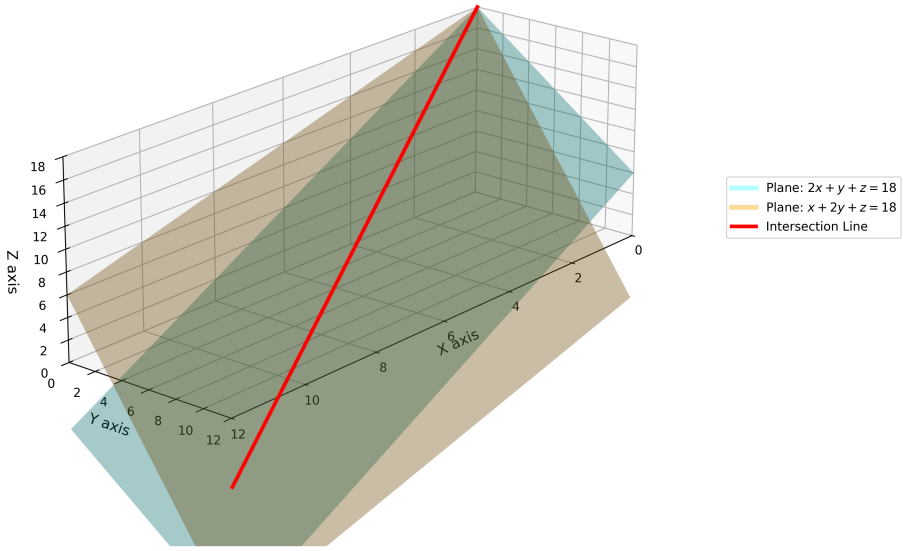


Figure 3: The constraint set

(i) The Lagrangian function is

$$L(x, y, z, \mu_1, \mu_2) = xyz - \mu_1((2x + y + z) - 18) - \mu_2((x + 2y + z) - 18).$$

The FOC are given by

$$\begin{cases} L_x = 0 \\ L_y = 0 \\ L_z = 0 \\ L_{\mu_1} = 0 \\ L_{\mu_2} = 0 \end{cases} \Rightarrow \begin{cases} yz - 2\mu_1 - \mu_2 = 0, & \cdots (1) \\ xz - \mu_1 - 2\mu_2 = 0, & \cdots (2) \\ xy - \mu_1 - \mu_2 = 0, & \cdots (3) \\ 2x + y + z = 18, & \cdots (4) \\ x + 2y + z = 18, & \cdots (5) \end{cases}$$

(ii) Let $\mathbf{p} = (x, y, z)$ be any point from the constraint set. Then its Jacobian matrix is

$$D\mathbf{h}(\mathbf{p}) = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix} \xrightarrow{\text{REF}} \begin{pmatrix} 1 & 2 & 1 \\ 0 & -3 & -1 \end{pmatrix}$$

which is of rank 2. Therefore, $\mathbf{p}$ satisfies the NDCQ. Moreover, since $\mathbf{p}$ is

arbitrary, we conclude that NDCQ is valid for this constraint set.

(iii) Firstly, by subtracting (5) from (4), we have

$$(2x + y + z) - (x + 2y + z) = 18 - 18.$$

Hence,

$$x - y = 0, \qquad x = y.$$

Then by (4), we have

$$2x + x + z = 18.$$

Thus,

$$3x + z = 18, \qquad z = 18 - 3x.$$

Next, subtracting (2) from (1), we have

$$(yz - 2\mu_1 - \mu_2) - (xz - \mu_1 - 2\mu_2) = 0.$$

Since $x = y$, this becomes

$$xz - 2\mu_1 - \mu_2 - xz + \mu_1 + 2\mu_2 = 0.$$

Hence,

$$-\mu_1 + \mu_2 = 0, \qquad \mu_1 = \mu_2.$$

Let

$$\mu_1 = \mu_2 = \mu.$$

Then (3) becomes

$$xy - \mu_1 - \mu_2 = 0.$$

Since $x = y$, we have

$$x^2 - 2\mu = 0.$$

Hence,

$$\mu = \frac{x^2}{2}.$$

Also, (1) becomes

$$xz - 2\mu - \mu = 0.$$

Hence,

$$xz - 3\mu = 0.$$

Putting $\mu = \dfrac{x^2}{2}$ into this equation, we have

$$xz - \frac{3x^2}{2} = 0.$$

Thus,

$$x\left(z - \frac{3x}{2}\right) = 0.$$

- If $x = 0$, then $y = 0$ and by $z = 18 - 3x$, we have

  $$z = 18.$$

  Also,

  $$\mu_1 = \mu_2 = \frac{x^2}{2} = 0.$$

  Therefore,

  $$(x, y, z, \mu_1, \mu_2) = (0, 0, 18, 0, 0), \text{ at which } f = 0.$$

- If $z - \dfrac{3x}{2} = 0$, then

  $$z = \frac{3x}{2}.$$

  Since $z = 18 - 3x$, we have

  $$18 - 3x = \frac{3x}{2}.$$

  Hence,

  $$36 - 6x = 3x, \qquad 9x = 36, \qquad x = 4.$$

  Therefore,

  $$y = 4, \qquad z = 18 - 3(4) = 6.$$

  Also,

  $$\mu_1 = \mu_2 = \frac{x^2}{2} = \frac{4^2}{2} = 8.$$

Therefore,

$$(x, y, z, \mu_1, \mu_2) = (4, 4, 6, 8, 8), \text{ at which } f = 96.$$

Thus, we obtain two critical points

$$(x, y, z, \mu_1, \mu_2) = (0, 0, 18, 0, 0), \ (4, 4, 6, 8, 8).$$

From the constraints,

$$2x + y + z = 18, \qquad x + 2y + z = 18.$$

Subtracting the second equation from the first equation gives

$$x - y = 0,$$

so

$$x = y.$$

Then, substituting $y = x$ into $2x + y + z = 18$, we get

$$2x + x + z = 18.$$

Hence,

$$3x + z = 18,$$

and therefore

$$z = 18 - 3x.$$

So the objective function becomes

$$f(x, y, z) = xyz = x \cdot x \cdot (18 - 3x).$$

Thus,

$$f(x, y, z) = x^2(18 - 3x) = 18x^2 - 3x^3.$$

As $x \to -\infty$,

$$18x^2 - 3x^3 \to +\infty.$$

Therefore, the objective function is unbounded above on the constraint set.

For example, take

$$x = y = -10, \qquad z = 48.$$

Then

$$2(-10) + (-10) + 48 = 18,$$

and

$$(-10) + 2(-10) + 48 = 18,$$

so this point satisfies both constraints. At this point,

$$f(x, y, z) = (-10)(-10)(48) = 4800 > 96.$$

Therefore,

$$\boxed{\text{There is no global maximum value.}}$$

**Problem 2**

(a) We first note that the given constraints define a closed and bounded subsets of $\mathbb{R}^2$. (See Figure 4 below)

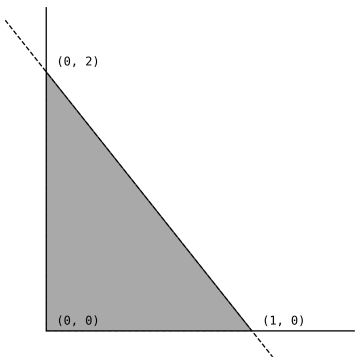


Figure 4: The constraint set

and hence by *Extreme Value Theorem*, the function $f(x, y)$ attains its maximum value on this constraint set. We rewrite the last two constraints as $-x \leq 0$ and $-y \leq 0$.

(i) The Lagrangian function is

$$L(x, y, \lambda_1, \lambda_2, \lambda_3) = x^2 + y^2 - \lambda_1((2x + y) - 2) - \lambda_2(-x) - \lambda_3(-y).$$

The FOC are given by

$$\begin{cases} L_x = 2x - 2\lambda_1 + \lambda_2 = 0, & \cdots (1) \\ L_y = 2y - \lambda_1 + \lambda_3 = 0, & \cdots (2) \\ \lambda_1(2x + y - 2) = 0, & \cdots (3) \\ \lambda_2(-x) = 0, & \cdots (4) \\ \lambda_3(-y) = 0, & \cdots (5) \\ 2x + y \leq 2,\ x \geq 0,\ y \geq 0, & \cdots (6) \\ \lambda_1 \geq 0,\ \lambda_2 \geq 0,\ \lambda_3 \geq 0, & \cdots (7) \end{cases}$$

(ii) We discuss the following cases.

$\boxed{\text{Case 1. Only one constraint is binding.}}$

The possible Jacobian matrices are

$$\begin{pmatrix} 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \end{pmatrix}.$$

All of the above matrices have rank 1.

$\boxed{\text{Case 2. Exactly two constraints are binding.}}$

The possible Jacobian matrices are

$$\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

All of the above matrices have rank 2.

$\boxed{\text{Case 3. All constraints are binding.}}$

Then we have $2x + y = 2$, $x = 0$, and $y = 0$ which is clearly impossible so the case is rejected.

Therefore, in all possible cases, the Jacobian matrix has rank that equals the number of binding constraint(s). This means that the NDCQ is satisfied.

(iii) Suppose $\lambda_1 = 0$. Then (1) becomes

$$2x + \lambda_2 = 0.$$

Since by (6) and (7) we have $x \geq 0$ and $\lambda_2 \geq 0$, the previous equality forces

$$x = \lambda_2 = 0.$$

Similarly, (2) becomes

$$2y + \lambda_3 = 0.$$

Since $y \geq 0$ and $\lambda_3 \geq 0$, we get

$$y = \lambda_3 = 0.$$

Thus, we obtain

$$(x, y, \lambda_1, \lambda_2, \lambda_3) = (0, 0, 0, 0, 0),$$

at which $f = 0$.

Suppose $\lambda_1 \neq 0$. Then by (7), $\lambda_1 > 0$, so (3) implies

$$2x + y = 2. \tag{3'}$$

Also, (4) implies $\lambda_2 = 0$ or $x = 0$, and (5) implies $\lambda_3 = 0$ or $y = 0$.

- If $\lambda_2 = 0$ and $\lambda_3 = 0$, then (1) and (2) become

  $$2x - 2\lambda_1 = 0, \qquad 2y - \lambda_1 = 0.$$

  Hence

  $$x = \lambda_1, \qquad 2y = \lambda_1.$$

  Therefore $x = 2y$. Substituting this into (3') gives

  $$2(2y) + y = 2,$$

  so

  $$y = \frac{2}{5}, \qquad x = \frac{4}{5}.$$

  Then $\lambda_1 = x = \frac{4}{5}$. Thus, we obtain

  $$(x, y, \lambda_1, \lambda_2, \lambda_3) = \left(\frac{4}{5}, \frac{2}{5}, \frac{4}{5}, 0, 0\right),$$

  at which $f = \frac{4}{5}$

- If $\lambda_2 = 0$ and $y = 0$, then (3') gives

$$2x = 2,$$

so $x = 1$. Then (1) gives

$$2 - 2\lambda_1 = 0,$$

so $\lambda_1 = 1$. Also, (2) gives

$$-\lambda_1 + \lambda_3 = 0,$$

so $\lambda_3 = 1$. Thus, we obtain

$$(x, y, \lambda_1, \lambda_2, \lambda_3) = (1, 0, 1, 0, 1),$$

at which $f = 1$.

- If $x = 0$ and $\lambda_3 = 0$, then (3') gives

$$y = 2.$$

Then (2) gives

$$4 - \lambda_1 = 0,$$

so $\lambda_1 = 4$. Also, (1) gives

$$-2\lambda_1 + \lambda_2 = 0,$$

so $\lambda_2 = 8$. Thus, we obtain

$$(x, y, \lambda_1, \lambda_2, \lambda_3) = (0, 2, 4, 8, 0),$$

at which $f = 4$.

- If $x = 0$ and $y = 0$, then (3') gives

$$0 = 2,$$

which is impossible.

Thus, we obtain four critical points

$$(x, y, \lambda_1, \lambda_2, \lambda_3) = (0, 0, 0, 0, 0)\,,\ \left(\frac{4}{5}, \frac{2}{5}, \frac{4}{5}, 0, 0\right),$$

$$(1, 0, 1, 0, 1)\,,\ (0, 2, 4, 8, 0)\,.$$

The maximum value of $f$ is 4.

(b) We first note that the given constraints define a closed and bounded subsets of $\mathbb{R}^3$. (See Figure 5 below)

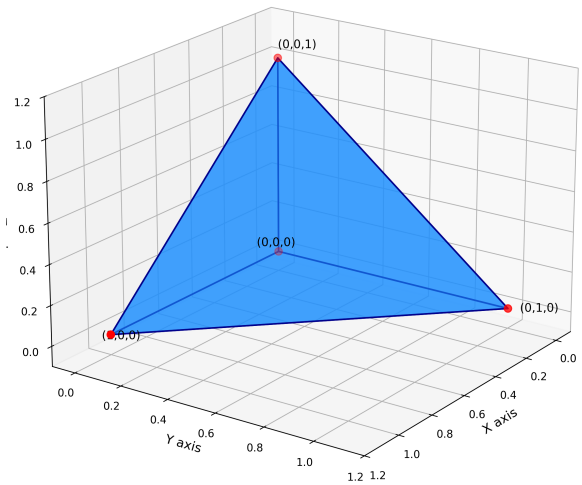


Figure 5: The constraint set

and hence by *Extreme Value Theorem*, the function $f(x, y)$ attains its maximum value on this constraint set. We rewrite the last three constraints as $-x \leq 0$, $-y \leq 0$, and $-z \leq 0$.

(i) The Lagrangian function is

$$L(x, y, z, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = xyz - \lambda_1((x + y + z) - 1) - \lambda_2(-x) - \lambda_3(-y) - \lambda_4(-z).$$

The FOC are given by

$$\begin{cases} L_x = yz - \lambda_1 + \lambda_2 = 0, & \cdots (1) \\ L_y = xz - \lambda_1 + \lambda_3 = 0, & \cdots (2) \\ L_z = xy - \lambda_1 + \lambda_4 = 0, & \cdots (3) \\ \lambda_1(x + y + z - 1) = 0, & \cdots (4) \\ \lambda_2(-x) = 0, & \cdots (5) \\ \lambda_3(-y) = 0, & \cdots (6) \\ \lambda_4(-z) = 0, & \cdots (7) \\ x + y + z \leq 1,\ x \geq 0,\ y \geq 0,\ z \geq 0, & \cdots (8) \\ \lambda_1 \geq 0,\ \lambda_2 \geq 0,\ \lambda_3 \geq 0,\ \lambda_4 \geq 0 & \cdots (9) \end{cases}$$

(ii) We discuss the following cases.

$\boxed{\text{Case 1. Only one constraint is binding.}}$

The possible Jacobian matrices are

$$\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}.$$

All of the above matrices have rank 1.

$\boxed{\text{Case 2. Exactly two constraints are binding.}}$

The possible Jacobian matrices are

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

All of the above matrices have rank 2.

$\boxed{\text{Case 3. Exactly three constraints are binding.}}$

The possible Jacobian matrices are

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

All of the above matrices have rank 3.

$\boxed{\text{Case 4. All constraints are binding.}}$

Then we have $x+y+z=1$, $x=0$, $y=0$, and $z=0$ which is clearly impossible so the case is rejected.

Therefore, in all possible cases, the Jacobian matrix has rank that equals the number of binding constraint(s). This means that the NDCQ is satisfied.

(iii) Suppose $\lambda_1 = 0$. Then (1), (2), and (3) become

$$yz + \lambda_2 = 0, \qquad xz + \lambda_3 = 0, \qquad xy + \lambda_4 = 0.$$

Since by (8) and (9) we have

$$x \geq 0, \quad y \geq 0, \quad z \geq 0, \qquad \lambda_2, \lambda_3, \lambda_4 \geq 0,$$

the previous equalities force

$$yz = 0, \qquad xz = 0, \qquad xy = 0,$$

and

$$\lambda_2 = \lambda_3 = \lambda_4 = 0.$$

Thus at most one of $x, y, z$ can be nonzero. Together with (8), we obtain

$$(x, y, z) = (t, 0, 0), \quad (0, t, 0), \quad \text{or} \quad (0, 0, t), \qquad 0 \leq t \leq 1.$$

Hence, in this case,

$$(x, y, z, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = (t, 0, 0, 0, 0, 0, 0),$$

or

$$(x, y, z, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = (0, t, 0, 0, 0, 0, 0),$$

or

$$(x, y, z, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = (0, 0, t, 0, 0, 0, 0),$$

where $0 \leq t \leq 1$.

Suppose $\lambda_1 \neq 0$. Then by (9), $\lambda_1 > 0$. Therefore, the complementary slackness condition gives

$$x + y + z = 1. \tag{4'}$$

We claim that $x, y, z > 0$. Indeed, suppose $x = 0$. Since $x+y+z=1$, at least

one of $y, z$ is positive.

If $y > 0$, then (6) implies $\lambda_3 = 0$. But (2) becomes

$$-\lambda_1 + \lambda_3 = 0,$$

so $\lambda_1 = 0$, a contradiction.

If $z > 0$, then (7) implies $\lambda_4 = 0$. But (3) becomes

$$-\lambda_1 + \lambda_4 = 0,$$

so $\lambda_1 = 0$, again a contradiction.

Hence $x \neq 0$. Similarly, we must have

$$y \neq 0, \qquad z \neq 0.$$

Therefore,

$$x > 0, \qquad y > 0, \qquad z > 0.$$

By (5), (6), and (7), this implies

$$\lambda_2 = \lambda_3 = \lambda_4 = 0.$$

Then (1), (2), and (3) become

$$yz - \lambda_1 = 0, \qquad xz - \lambda_1 = 0, \qquad xy - \lambda_1 = 0.$$

Thus,

$$yz = xz = xy.$$

Since $x, y, z > 0$, we get

$$x = y = z.$$

Together with (4'), this gives

$$3x = 1,$$

so

$$x = y = z = \frac{1}{3}.$$

Also,

$$\lambda_1 = yz = \frac{1}{9}.$$

Thus, we obtain one critical point

$$(x, y, z, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, \frac{1}{9}, 0, 0, 0\right),$$

at which $f = \frac{1}{27}$.

Hence, $f = \frac{1}{27}$ is the maximum value.

(c) We first note that the given constraints define a closed and bounded subsets of $\mathbb{R}^4$, and hence by *Extreme Value Theorem*, the function $f(x, y)$ attains its maximum value on this constraint set.

(i) The Lagrangian function is

$$L(x, y, z, w, \lambda_1, \lambda_2) = xz + yw - \lambda_1((x^2 + y^2) - 1) - \lambda_2((4z^2 + 9w^2) - 36).$$

The FOC are given by

$$\begin{cases} L_x = z - 2\lambda_1 x = 0, & \cdots (1) \\ L_y = w - 2\lambda_1 y = 0, & \cdots (2) \\ L_z = x - 8\lambda_2 z = 0, & \cdots (3) \\ L_w = y - 18\lambda_2 w = 0, & \cdots (4) \\ \lambda_1(x^2 + y^2 - 1) = 0, & \cdots (5) \\ \lambda_2(4z^2 + 9w^2 - 36) = 0, & \cdots (6) \\ x^2 + y^2 \leq 1,\ 4z^2 + 9w^2 \leq 36 & \cdots (7) \\ \lambda_1 \geq 0,\ \lambda_2 \geq 0 & \cdots (8) \end{cases}$$

(ii) We discuss the following cases.

$\boxed{\text{Case 1. Only one constraint is binding.}}$

The possible Jacobian matrices are

$$\begin{pmatrix} 2x & 2y & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 8z & 18w \end{pmatrix}.$$

Note that if $2x = 2y = 0$, then $x = y = 0$ so the constraint $x^2 + y^2 \leq 1$ cannot

be binding. Similarly, if $8z = 18w = 0$, then $z = w = 0$ so the constraint $4z^2 + 9w^2 \leq 36$ cannot be binding. Therefore, all of the above matrices have rank 1.

$\boxed{\text{Case 2. All (two) constraints are binding.}}$

The possible Jacobian matrix is

$$\begin{pmatrix} 2x & 2y & 0 & 0 \\ 0 & 0 & 8z & 18w \end{pmatrix}.$$

If the above matrix has rank $< 2$, it requires $2x = 2y = 0$ or $8z = 18w = 0$. However, we have shown in Case 1 that this would violate the binding of both constraints. Therefore, the above matrix has rank 2.

Therefore, in all possible cases, the Jacobian matrix has rank that equals the number of binding constraint(s). This means that the NDCQ is satisfied.

(iii) Suppose $\lambda_1 = 0$. Then (1) and (2) become

$$z = 0, \qquad w = 0.$$

Then (3) and (4) imply

$$x = 0, \qquad y = 0.$$

Also, (6) becomes

$$\lambda_2(0 - 36) = 0,$$

so $\lambda_2 = 0$. Thus, we obtain

$$(x, y, z, w, \lambda_1, \lambda_2) = (0, 0, 0, 0, 0, 0),$$

at which $f = 0$.

Suppose $\lambda_1 \neq 0$. Then by (8), $\lambda_1 > 0$, so (5) implies

$$x^2 + y^2 = 1. \tag{5'}$$

In particular, $(x, y) \neq (0, 0)$.

If $\lambda_2 = 0$, then (3) and (4) imply

$$x = 0, \qquad y = 0,$$

which contradicts (5′). Hence we must have

$$\lambda_2 > 0.$$

Therefore, (6) implies

$$4z^2 + 9w^2 = 36. \tag{6'}$$

From (1) and (3), we have

$$z = 2\lambda_1 x, \qquad x = 8\lambda_2 z.$$

Substituting the first equation into the second gives

$$x = 8\lambda_2(2\lambda_1 x) = 16\lambda_1\lambda_2 x.$$

Hence

$$(1 - 16\lambda_1\lambda_2)x = 0. \tag{9}$$

Similarly, from (2) and (4), we have

$$w = 2\lambda_1 y, \qquad y = 18\lambda_2 w.$$

Substituting the first equation into the second gives

$$y = 18\lambda_2(2\lambda_1 y) = 36\lambda_1\lambda_2 y.$$

Hence

$$(1 - 36\lambda_1\lambda_2)y = 0. \tag{10}$$

If $x \neq 0$ and $y \neq 0$, then (9) and (10) imply

$$16\lambda_1\lambda_2 = 1, \qquad 36\lambda_1\lambda_2 = 1,$$

which is impossible. Therefore, at least one of $x, y$ must be zero.

- If $x = 0$, then by (5′) we have

$$y = \pm 1.$$

From (1), we get

$$z = 0.$$

Then (6′) gives

$$9w^2 = 36,$$

so

$$w = \pm 2.$$

Since (2) gives

$$w = 2\lambda_1 y,$$

and $\lambda_1 > 0$, $w$ and $y$ must have the same sign. Hence

$$(y, w) = (1, 2) \quad \text{or} \quad (y, w) = (-1, -2).$$

If $(y, w) = (1, 2)$, then

$$2 = 2\lambda_1,$$

so $\lambda_1 = 1$. Also, (4) gives

$$1 = 18\lambda_2(2),$$

so

$$\lambda_2 = \frac{1}{36}.$$

Thus, we obtain

$$(x, y, z, w, \lambda_1, \lambda_2) = \left(0, 1, 0, 2, 1, \frac{1}{36}\right),$$

at which $f = 2$.

Similarly, if $(y, w) = (-1, -2)$, we obtain

$$(x, y, z, w, \lambda_1, \lambda_2) = \left(0, -1, 0, -2, 1, \frac{1}{36}\right),$$

at which $f = 2$.

- If $y = 0$, then by (5′) we have

$$x = \pm 1.$$

From (2), we get

$$w = 0.$$

Then (6′) gives

$$4z^2 = 36,$$

so

$$z = \pm 3.$$

Since (1) gives

$$z = 2\lambda_1 x,$$

and $\lambda_1 > 0$, $z$ and $x$ must have the same sign. Hence

$$(x, z) = (1, 3) \quad \text{or} \quad (x, z) = (-1, -3).$$

If $(x, z) = (1, 3)$, then

$$3 = 2\lambda_1,$$

so

$$\lambda_1 = \frac{3}{2}.$$

Also, (3) gives

$$1 = 8\lambda_2(3),$$

so

$$\lambda_2 = \frac{1}{24}.$$

Thus, we obtain

$$(x, y, z, w, \lambda_1, \lambda_2) = \left(1, 0, 3, 0, \frac{3}{2}, \frac{1}{24}\right),$$

at which $f = 3$.

Similarly, if $(x, z) = (-1, -3)$, we obtain

$$(x, y, z, w, \lambda_1, \lambda_2) = \left(-1, 0, -3, 0, \frac{3}{2}, \frac{1}{24}\right),$$

at which $f = 3$.

Thus, we obtain five critical points

$$(x, y, z, w, \lambda_1, \lambda_2) = (0, 0, 0, 0),\ \left(0, \pm 1, 0, \pm 2, 1, \frac{1}{36}\right),\ \left(\pm 1, 0, \pm 3, 0, \frac{3}{2}, \frac{1}{24}\right).$$

The maximum value of $f$ is 3.